

Spring REST Exercise 1: REST Hello Endpoint

Objectives

Demonstrate creation of Spring Boot Application

Spring initializr, <https://start.spring.io>, @SpringBootApplication, SpringApplication.run()

Ref - <https://start.spring.io>

Explain the need and benefits of Spring Boot

Makes Java development easy, avoids tedious development steps, reduces development time, avoids writing boilerplate code, provides embedded tomcat server, avoid XML configuration

Ref - <https://www.journaldev.com/7969/spring-boot-tutorial>

Demonstrate loading bean from spring configuration file

Spring configuration xml, spring xml schema spring-beans.xsd, <bean>, id, class, <constructor-arg>, <property>, name, value, ClassPathXmlApplicationContext, ApplicationContext, context.getBean(), singleton scope, prototype scope

Ref - <https://docs.spring.io/spring-framework/docs/current/spring-framework-reference/core.html>

IoC	Container	-
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<https://docs.spring.io/spring-framework/docs/current/spring-framework-reference/core.html#beans>

Scopes	-
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<https://docs.spring.io/spring-framework/docs/current/spring-framework-reference/core.html#beans-factory-scopes>

Constructor	Injection	-
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<https://docs.spring.io/spring-framework/docs/current/spring-framework-reference/core.html#beans-constructor-injection>

Setter	method	injection	-
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<https://docs.spring.io/spring-framework/docs/current/spring-framework-reference/core.html#beans-setter-injection>

Demonstrate inclusion of logging in Spring Boot Application

application.properties, logging.level, logging.pattern, server.port, LoggerFactory, Logger, log levels (trace, debug, info, warn, error)

Ref - <https://docs.spring.io/spring-boot/docs/current/reference/html/boot-features-logging.html>

Hands on 1

Create a Spring Web Project using Maven

Follow steps below to create a project:

G- to <https://start.spring.io/>

Change Group as "com.cognizant"

Change Artifact Id as "spring-learn"

Select Spring Boot DevTools and Spring Web

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Create and download the project as zip

Extract the zip in root folder t- Eclipse Workspace

Build the project using `?mvn clean package -Dhttp.proxyHost=proxy.cognizant.com -Dhttp.proxyPort=6050 -Dhttps.proxyHost=proxy.cognizant.com -Dhttps.proxyPort=6050 -Dhttp.proxyUser=123456'` command in command line

Import the project in Eclipse "File > Import > Maven > Existing Maven Projects > Click Browse and select extracted folder > Finish"

Include logs t- verify if main() method of SpringLearnApplication.

Run the SpringLearnApplication class.

SME t- walk through the following aspects related t- the project created:

src/main/java - Folder with application code

src/main/resources - Folder for application configuration

src/test/java - Folder with code for testing the application

SpringLearnApplication.java - Walkthrough the main() method.

Purpose of @SpringBootApplication annotation

pom.xml

Walkthrough all the configuration defined in XML file

Open 'Dependency Hierarchy' and show the dependency tree.

Hands on 2

Spring Core - Load SimpleDateFormat from Spring Configuration XML

SimpleDateFormat with the pattern `'dd/MM/yyyy'` is created in multiple places of an application. T- avoid creation of SimpleDateFormat in multiple places, define a bean in Spring XML Configuration file and retrieve the date.

Follow steps below t- implement:

Create spring configuration file date-format.xml in src/main/resources folder of 'spring-learn' project

Open <https://docs.spring.io/spring-framework/docs/current/spring-framework-reference/core.html#beans-factory-metadata>

Copy the XML defined in the section of previous step URL and paste it into date-format.xml

Define bean tag in the XML with for date format. Refer code below.

```
<?xml version="1.0" encoding="UTF-8"?>
```

```
<beans xmlns="http://www.springframework.org/schema/beans"
```

```
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
```

```
xsi:schemaLocation="http://www.springframework.org/schema/beans
```

```
https://www.springframework.org/schema/beans/spring-beans.xsd">
```

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```
<bean id="dateFormat" class="java.text.SimpleDateFormat">
<constructor-arg value="dd/MM/yyyy" />
</bean>
</beans>
```

Create new method displayDate() in SpringLearnApplication.java

In displayDate() method create the ApplicationContext. Refer code below:

```
ApplicationContext context = new ClassPathXmlApplicationContext("date-format.xml");
```

Get the dateFormat using getBean() method. Refer code below.

```
SimpleDateFormat format = context.getBean("dateFormat", SimpleDateFormat.class);
```

Using the format variable try to parse string '31/12/2018' to Date class and display the result using System.out.println.

Run the application as 'Java Application' and check the result in console log output.

Troubleshooting Tips

If the tomcat port has a conflict and the server is not starting include the below property in application.properties file in src/main/resources folder.

Hands on 3

Spring Core - Incorporate Logging

Incorporate logging in the Spring Boot project created in previous hands on. Refer steps below:

Create application.properties if not yet created in src/main/resources folder

Add below lines in application.properties

```
logging.level.org.springframework=info
```

```
logging.level.com.cognizant.springlearn=debug
```

```
logging.pattern.console=%d{yyMMdd}|%d{HH:mm:ss.SSS}|%-20.20thread|%-5p|%-25.25logger{25}|%-25M|%m%n
```

In SpringLearnApplication.java include the following imports:

```
import org.slf4j.Logger;
```

```
import org.slf4j.LoggerFactory;
```

Include the below static variable in SpringLearnApplication.java:

```
private static final Logger LOGGER = LoggerFactory.getLogger(SpringLearnApplication.class);
```

Include info- log on start and end of method. Debug log for displaying the date (refer code below)

```
public void displayDate() {
```

```
    LOGGER.info("START");
```

```
    //..
```

```
    LOGGER.debug(date);
```

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```
//..  
LOGGER.info("END");  
}
```

IMPORTANT NOTE: Going forward all methods should incorporate logging as specified above. Never use `System.out.println()`.

Hands on 4

Spring Core - Load Country from Spring Configuration XML

An airlines website is going to support booking on four countries. There will be a drop down on the home page of this website to select the respective country. It is also important to store the two-character ISO code of each country.

Above data has to be stored in spring configuration file. Write a program to read this configuration file and display the details.

Steps to implement

Pick any one of your choice country to configure in Spring XML configuration named `country.xml`.

Create a bean tag in spring configuration for country and set the property and values

```
<bean id="country" class="com.cognizant.springlearn.Country">  
<property name="code" value="IN" />  
<property name="name" value="India" />  
</bean>
```

Create Country class with following aspects:

Instance variables for code and name

Implement empty parameter constructor with inclusion of debug log within the constructor with log message as "Inside Country Constructor."

Generate getters and setters with inclusion of debug with relevant message within each setter and getter method.

Generate `toString()` method

Create a method `displayCountry()` in `SpringLearnApplication.java`, which will read the country bean from spring configuration file and display the country details. `ClassPathXmlApplicationContext`, `ApplicationContext` and `context.getBean("beanId", Country.class)`. Refer sample code for `displayCountry()` method below.

```
ApplicationContext context = new ClassPathXmlApplicationContext("country.xml");
```

```
Country country = (Country) context.getBean("country", Country.class);
```

```
LOGGER.debug("Country : {}", country.toString());
```

Invoke `displayCountry()` method in `main()` method of `SpringLearnApplication.java`.

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Execute main() method and check the logs to find out which constructors and methods were invoked.

SME to provide more detailing about the following aspects:

bean tag, id attribute, class attribute, property tag, name attribute, value attribute

ApplicationContext, ClassPathXmlApplicationContext

What exactly happens when context.getBean() is invoked

Hands on 5

Spring Core - Demonstration of Singleton Scope and Prototype Scope

The Country bean done in the previous hands on will be used to demonstrate the scopes in Spring. Implement the steps below.

Follow steps below to demonstrate Singleton Scope

Include a line in displayCountry() to get country bean reference one more time from the same application context. Only the third line of the below code snippet should be copied and pasted.

```
ApplicationContext context = new ClassPathXmlApplicationContext("country.xml");
```

```
Country country = context.getBean("country", Country.class);
```

```
Country anotherCountry = context.getBean("country", Country.class);
```

The constructor will be called only once, which means that only one instance of Country bean is created

Follow steps below to demonstrate Prototype Scope

Include scope="prototype" attribute in bean definition xml.

```
<bean id="country" class="com.cognizant.springlearn.Country" scope="prototype">
```

Run the application

Constructor will be called twice, which means that two instances of country are created.

Hands on 6

Spring Core - Load list of countries from Spring Configuration XML

Our main objective was to retrieve the list of four countries for the airlines website. Refer steps below to get this incorporated.

Create a separate bean for each of the four countries in country.xml.

Create an ArrayList of Country in country.xml. Refer code below.

```
<bean id="countryList" class="java.util.ArrayList">
```

```
<constructor-arg>
```

```
<list>
```

```
<ref bean="in"></ref>
```

```
<ref bean="us"></ref>
```

```
<ref bean="de"></ref>
```

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<ref bean="jp"></ref>

</list>

</constructor-arg>

</bean>

Include new method displayCountries() in SpringLearnApplication.java

In displayCountries() read the country list created above

Display the list of countries as debug log.

SME t- provide detailing on below aspects:

<list>

<ref>

bean attribute

IMPORTANT NOTE: D- not forget t- include the start and end logs in this new method.

Code | Name

US | United States

DE | Germany

IN | India

JP | Japan